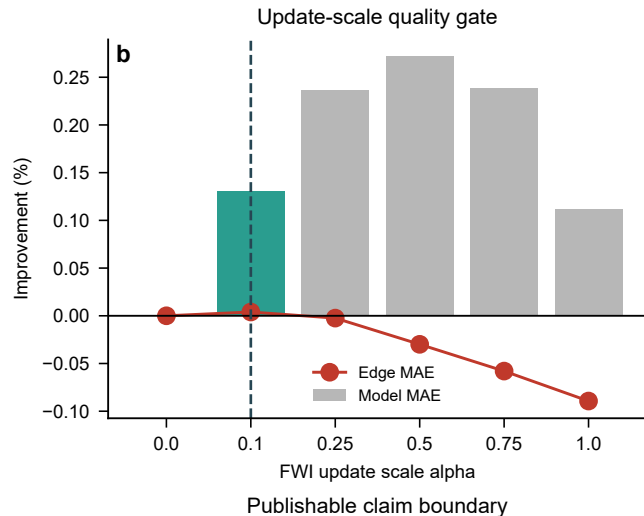


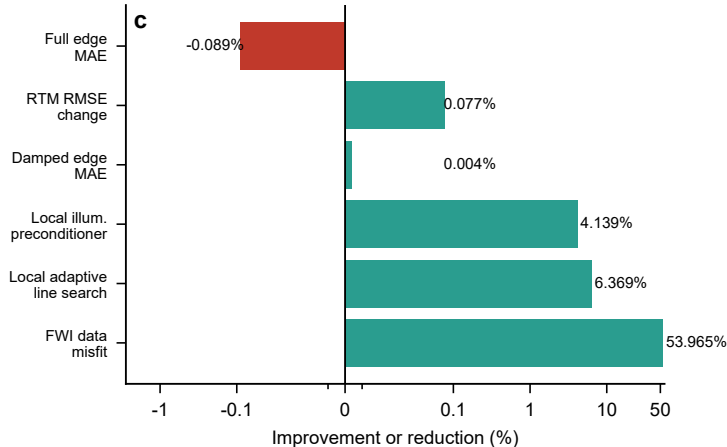
Literature-guided method synthesis

a

Literature need	Risk in this result	Implemented response
FWI challenge	cycle skipping; weak salt edges	model-quality gate
Regularization	structure preservation	edge/gradient metrics
Optimization	step-length sensitivity	alpha scan + line search
RTM/LSRTM	image condition bias	RTM before/after validation
Benchmarks	reproducibility	SEG/Salt + fixed scripts



Evidence strength and limitation



d

High-quality FWI velocity recovery

No

Quality-gated FWI-to-RTM workflow

Yes

RTM imaging-condition diagnostics

Yes

Illumination scaling as primary FWI optimizer

No

ML/foundation-model prior as main result

Future